

Dyadic Spectral Divisors and Gamma Duality

New Laguerre–Pólya, non-holonomic, heat-trace, and Thorin frontiers for
the Fabius and Rvachev functions

Research report prepared from the `ProveIt` documentation corpus

30 August 2026

Abstract

The current `ProveIt` corpus already contains a very broad formal and semi-formal theory of the Fabius function, its inverse, Rvachev’s compactly supported up-function, Thue–Morse identities, dyadic arithmetic, geometric- q laws, infinite sinc products, Lambert- W endpoint asymptotics, Legendre expansions, and Bell–Bernoulli moment algebra. This report deliberately starts from a narrow piece of that established theory—the exact integer-zero divisor of the Rvachev Fourier product—and follows consequences that are absent from the current primary and frontier volumes.

The main new results are as follows. First, the Rvachev Fourier transform is placed in the Laguerre–Pólya class, while a square-root moment transform belongs to the type-I subclass. Its normalized even moments therefore form a Pólya-frequency sequence of infinite order, all shifted Jensen polynomials are hyperbolic, every derivative of the Fourier transform has only real zeros, and the moments satisfy a new sharp two-sided Turán hierarchy. Second, the unbounded dyadic zero multiplicities prove that the Fourier transform and its moment transform are not D-finite; consequently the exact even-moment sequence is not P-recursive. Third, reciprocal-integer bases are treated uniformly: the products are Cartwright-class entire functions whose exact zero counts contain base- b digit sums.

A second chain of results gives a probabilistic spectral dual. The logarithmic derivative of the moment transform is a Stieltjes transform, its logarithm is a complete Bernstein function, and every negative power is the Laplace transform of an explicit generalized gamma convolution with a purely atomic Thorin measure. Its Lévy density is controlled by an arithmetic heat trace. Mellin inversion gives the heat trace’s complete algebraic and log-periodic skeleton. Jacobi inversion then supplies an *exact* beyond-all-orders remainder, not merely an asymptotic estimate. Laplace transformation of this identity yields an exact modular-dual formula for the infinite sinc moment product; the exponentially small factor is the generating product for partitions into powers of b . The leading heat defect has an explicit lower-Lambert- W inverse, creating a new bridge to the inverse-Fabius asymptotic machinery.

All claims marked “Theorem” or “Corollary” are proved in the report. Conjectures are isolated in a separate section. A commented Python program reproduces exact moment tables, Jensen-root checks, periodic Fourier coefficients, and the heat/modular-dual numerical validations.

Contents

1	Corpus audit, scope, and novelty boundary	3
1.1	Documents examined	3
1.2	Established inputs reused here	3
1.3	Non-duplication matrix	4

2	Normalization and reciprocal-integer-base family	4
2.1	The dyadic law and its base- b analogue	4
2.2	The square-root moment transform	5
3	Laguerre–Pólya structure, total positivity, and Jensen hyperbolicity	5
3.1	Real-rooted entire functions	5
3.2	Pólya-frequency moments	6
3.3	All shifted Jensen polynomials	6
4	Bernoulli, Bell, and symmetric-function formulas	8
4.1	Spectral power sums	8
4.2	Newton recurrence and Bell polynomial	8
5	Non-holonomicity and non-P-recursive moments	10
5.1	A zero-multiplicity obstruction	10
6	Cartwright class and digital zero counts	10
7	Complete Bernstein structure and a Thorin gamma dual	11
7.1	Stieltjes logarithmic derivative	11
7.2	Generalized gamma convolution	12
8	The arithmetic heat trace	13
8.1	Definition, q -renewal, and Malmsten representation	13
8.2	Mellin transform and log-periodic skeleton	13
8.3	Exact beyond-all-orders remainder	15
9	Exact modular duality for the infinite sinc moment product	16
9.1	Laplace transform of the heat identity	16
9.2	A partition product hidden in the sinc product	18
10	Consequences for computation and approximation	19
10.1	Certified Stieltjes–Padé control	19
10.2	Moment cones and Legendre expansions	19
10.3	Inverse-Fabius transfer	19
11	Conjectures and frontier problems	20
12	Summary of the new theorem network	21
A	Reproducibility	21
B	Selected numerical constants	22

1 Corpus audit, scope, and novelty boundary

1.1 Documents examined

The source audit used the live documentation tree under [12]

<https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs>.

The principal TeX artifacts are:

1. `Fabius_Function_and_Rvachev_Up.tex`, the approximately ten-thousand-line primary exposition of the formally verified development;
2. `semi-formalized-research-frontiers.tex`, the approximately fifteen-thousand-line frontier volume that consolidates the former subdirectory notebooks into six thematic syntheses;
3. `FabiusFunction_Mathematical_Glossary.tex`, with its terminology and notation cross-walk;
4. `Non_Elementarity_of_the_Fabius_Function.tex`;
5. `fabius_lean_walkthrough.tex`;
6. the TeX audit artifacts, source-paper drafts, and archived predecessor reports used by the consolidation provenance map.

The consolidated frontier volume explicitly states that it merges every mathematical document formerly stored below its subdirectory into six non-duplicative thematic syntheses, while retaining provenance and a formalization boundary. Accordingly, the audit treated the consolidated volume as the authoritative mathematical union and used the archived files primarily to check provenance and avoid resurrecting superseded formulations. Here “new” means absent from that live corpus; it is not a claim of literature-wide priority without a dedicated historical priority study.

1.2 Established inputs reused here

Only the following repository results are taken as inputs; classical background on Fabius and Rvachev functions is in [7, 13, 3, 4].

1. The Fourier normalization

$$\Phi(\xi) = \widehat{\text{up}}(\xi) = \int_{-1}^1 e^{-2\pi i \xi x} \text{up}(x) \, dx = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{\pi \xi}{2^j}\right), \quad \text{sinc } z = \frac{\sin z}{z}. \quad (1)$$

2. The exact integer-zero multiplicity

$$\text{ord}_{\xi=m} \Phi = 1 + v_2(|m|), \quad m \in \mathbb{Z} \setminus \{0\}. \quad (2)$$

3. The probability model

$$X = \sum_{j=0}^{\infty} \frac{U_j}{2^{j+1}}, \quad U_j \stackrel{\text{iid}}{\sim} \text{Unif}[-1, 1], \quad (3)$$

whose density is up on $[-1, 1]$.

4. The repository’s moment/Bernoulli algebra, q-Pochhammer factorization, Thue–Morse lobe-sign theorem, endpoint asymptotics, and Lambert- W inverse machinery, used only for comparison and future-transfer remarks.

No theorem below is presented as formally verified in Lean. The report follows the repository’s own rule: a mathematical proof in a frontier document is not a promoted formal theorem.

1.3 Non-duplication matrix

Topic	Status in current corpus	Status in this report
Exact integer zeros and multiplicities of Φ	Formalized primary input	Reused, not claimed as new
Complex q-Pochhammer factorization of Φ	Formalized primary input	Reused for comparison
Thue–Morse sign on Fourier lobes	Formalized primary input	Recovered as a consistency corollary
Laguerre–Pólya class and derivative real-rootedness	No matching treatment found	New theorem chain
Pólya-frequency moments and Jensen hyperbolicity	No matching treatment found	New theorem chain
D-finiteness/P-recursiveness obstruction	Explicitly listed as frontier	New proof and generalization
Cartwright classification	No matching treatment found	New consequence of support and divisor
Complete Bernstein and Thorin/GGC duality	No matching treatment found	New probabilistic construction
Arithmetic heat trace and Malmsten integral	Explicitly listed as frontier	New exact Mellin/Jacobi theory
Modular-dual lacunary partition product	No matching treatment found	New exact identity

2 Normalization and reciprocal-integer-base family

2.1 The dyadic law and its base- b analogue

For an integer $b \geq 2$, define

$$\Phi_b(\xi) := \prod_{j=0}^{\infty} \operatorname{sinc}\left(\frac{\pi\xi}{b^j}\right). \quad (4)$$

This is the Fourier transform of

$$X_b := \sum_{j=0}^{\infty} \frac{U_j}{2b^j}, \quad \operatorname{supp}(X_b) = \left[-\frac{b}{2(b-1)}, \frac{b}{2(b-1)}\right]. \quad (5)$$

The dyadic case $b = 2$ is exactly (1) and (3).

Let

$$\nu_b(m) := \max\{k \geq 0 : b^k \mid m\}, \quad d_b(m) := 1 + \nu_b(m), \quad m \geq 1. \quad (6)$$

For composite b , ν_b is a whole-base divisibility exponent, not a prime-adic valuation.

Proposition 2.1 (Arithmetic canonical product). *For every integer $b \geq 2$,*

$$\Phi_b(z) = \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{m^2}\right)^{d_b(m)}, \quad (7)$$

locally uniformly on $\mathbb{C}$. Hence the nonzero zeros are precisely the signed integers and

$$\text{ord}_{z=\pm m} \Phi_b = d_b(m). \quad (8)$$

Proof. Euler’s product $\text{sinc}(\pi w) = \prod_{n \geq 1} (1 - w^2/n^2)$ gives

$$\Phi_b(z) = \prod_{j \geq 0} \prod_{n \geq 1} \left(1 - \frac{z^2}{b^{2j}n^2}\right).$$

A positive integer m occurs as $b^j n$ exactly for $j = 0, 1, \dots, \nu_b(m)$. Regrouping therefore gives (7). The regrouping is legitimate because

$$\sum_{m \geq 1} \frac{d_b(m)}{m^2} = \frac{\zeta(2)}{1 - b^{-2}} < \infty.$$

The multiplicity statement follows immediately. □

2.2 The square-root moment transform

Because Φ_b is even, the expression

$$A_b(z) := \Phi_b\left(\frac{i\sqrt{z}}{\pi}\right) \quad (9)$$

is single-valued and entire. Equivalently,

$$A_b(z) = \prod_{j=0}^{\infty} \frac{\sinh(\sqrt{z}/b^j)}{\sqrt{z}/b^j} \quad (10)$$

$$= \prod_{m=1}^{\infty} \left(1 + \frac{z}{\pi^2 m^2}\right)^{d_b(m)}. \quad (11)$$

For $b = 2$, if $\mu_{2n} := \mathbb{E}[X^{2n}]$, then

$$A_2(z) = \mathbb{E}e^{2\sqrt{z}X} = \sum_{n=0}^{\infty} a_n z^n, \quad a_n = \frac{4^n \mu_{2n}}{(2n)!}. \quad (12)$$

The normalization removes all powers of π : every a_n is rational.

3 Laguerre–Pólya structure, total positivity, and Jensen hyperbolicity

3.1 Real-rooted entire functions

Theorem 3.1 (Laguerre–Pólya classification). *For every integer $b \geq 2$:*

- (i) Φ_b belongs to the Laguerre–Pólya class $\mathcal{LP}$.
- (ii) A_b belongs to the type-I Laguerre–Pólya class $\mathcal{LP}\text{-I}$.

- (iii) Every derivative $\Phi_b^{(r)}$ and $A_b^{(r)}$ has only real zeros.
- (iv) The zeros of successive derivatives interlace in the usual weak sense; after common multiplicity factors are removed, the interlacing is strict.

Proof. Truncate (7) at $m \leq M$. The resulting polynomial has only real zeros, and the products converge locally uniformly because $\sum d_b(m)m^{-2} < \infty$. Thus Φ_b is a locally uniform limit of real-rooted polynomials and belongs to $\mathcal{LP}$; see [11] for the standard Laguerre–Pólya framework.

Likewise, truncations of (11) have only negative zeros. The summability of $d_b(m)/(\pi^2 m^2)$ puts A_b in the canonical type-I form. Closure of $\mathcal{LP}$ under differentiation proves the real-zero statement. Interlacing follows by applying Rolle’s theorem to finite truncations and passing to the locally uniform limit, with repeated zeros carried through their common factors. $\square$

Remark 3.2. The conclusion is stronger than the already known location and multiplicity of the zeros of Φ . It controls the zeros of *every* Fourier derivative. In particular, no derivative creates a nonreal zero pair.

3.2 Pólya-frequency moments

Recall that a sequence $(c_n)_{n \geq 0}$ is a Pólya-frequency sequence of infinite order if every minor of the Toeplitz matrix $(c_{j-i})_{i,j \geq 0}$, with $c_k = 0$ for $k < 0$, is nonnegative.

Theorem 3.3 (Total nonnegativity of normalized moments). *For every integer $b \geq 2$, the coefficient sequence*

$$A_b(z) = \sum_{n \geq 0} a_n^{(b)} z^n \quad (13)$$

belongs to PF_∞ . In particular, all Toeplitz minors are nonnegative and

$$(a_n^{(b)})^2 \geq a_{n-1}^{(b)} a_{n+1}^{(b)}. \quad (14)$$

Proof. The Aissen–Edrei–Schoenberg–Whitney characterization [1, 2] says that a nonnegative sequence is PF_∞ precisely when its generating function has the form

$$C z^m e^{\gamma z} \frac{\prod_i (1 + \alpha_i z)}{\prod_j (1 - \beta_j z)}, \quad \alpha_i, \beta_j, \gamma \geq 0, \quad \sum_i \alpha_i + \sum_j \beta_j < \infty.$$

Equation (11) has this form with $C = 1$, $m = 0$, $\gamma = 0$, no β_j , and with $d_b(m)$ copies of $\alpha = (\pi^2 m^2)^{-1}$. The required sum is finite. $\square$

Corollary 3.4 (Variation-diminishing moment transform). *Convolution by the sequence $(a_n^{(b)})$ cannot increase the number of sign changes of a finitely supported real sequence. Thus the normalized Rvachev moment kernel is a discrete variation-diminishing operator.*

3.3 All shifted Jensen polynomials

Put

$$\gamma_n^{(b)} := n! a_n^{(b)} = A_b^{(n)}(0) \quad (15)$$

and define the shifted Jensen polynomial

$$J_{d,n}^{(b)}(x) := \sum_{j=0}^d \binom{d}{j} \gamma_{n+j}^{(b)} x^j. \quad (16)$$

Theorem 3.5 (Jensen hyperbolicity hierarchy). *For all integers $b \geq 2$, $d \geq 1$, and $n \geq 0$, the polynomial $J_{d,n}^{(b)}$ has only real negative zeros.*

Proof. The derivative $A_b^{(n)}$ is in $\mathcal{LP}$ by Theorem 3.1. Jensen’s characterization of the Laguerre–Pólya class (in the modern Jensen-polynomial formulation, see [9]) therefore makes its degree- d Jensen polynomial hyperbolic. Formula (16) has strictly positive coefficients, so a real zero cannot be nonnegative. Hence every zero is negative. $\square$

Corollary 3.6 (Strong Turán inequality). *For $n \geq 1$,*

$$(a_n^{(b)})^2 > \frac{n+1}{n} a_{n-1}^{(b)} a_{n+1}^{(b)}. \quad (17)$$

In the dyadic case this becomes the new two-sided moment inequality

$$\frac{2n-1}{2n+1} < \frac{\mu_{2n}^2}{\mu_{2n-2}\mu_{2n+2}} < 1, \quad n \geq 1. \quad (18)$$

Proof. Apply the strict Laguerre inequality

$$(f'(0))^2 - f(0)f''(0) > 0$$

to $f = A_b^{(n-1)}$. Strictness holds because $A_b^{(n-1)}$ has real zeros and is not an exponential. In terms of $\gamma_n = n!a_n$, this is $\gamma_n^2 > \gamma_{n-1}\gamma_{n+1}$, which rearranges to (17).

Using (12), the lower inequality is exactly

$$\frac{\mu_{2n}^2}{\mu_{2n-2}\mu_{2n+2}} > \frac{n+1}{n} \frac{(2n)!^2}{(2n-2)!(2n+2)!} = \frac{2n-1}{2n+1}.$$

The upper inequality is the strict log-convexity of the moment sequence, obtained from Cauchy–Schwarz applied to $|X|^{n-1}$ and $|X|^{n+1}$. Equality would force $|X|$ to be almost surely constant, which is false for the up-density. $\square$

Numerical experiment 3.7 (Jensen roots). The accompanying program constructs exact rational coefficients and computes roots at 70 digits. The largest observed imaginary part in the grid $d \in \{2, 4, 6, 8, 10\}$, $n \in \{0, 3, 6\}$ is zero at the working precision.

degree	shift	max. $ \Im \rho $	nearest root to 0	leftmost root
2	0	0.00e+00	-3.02034	-8.82176
2	3	0.00e+00	-6.2162	-12.9274
2	6	0.00e+00	-9.16584	-16.7927
4	0	0.00e+00	-1.85499	-16.638
4	3	0.00e+00	-4.23958	-21.2172
4	6	0.00e+00	-6.57067	-25.5285
6	0	0.00e+00	-1.34429	-24.3828
6	3	0.00e+00	-3.25402	-29.2044
6	6	0.00e+00	-5.19761	-33.7791
8	0	0.00e+00	-1.05519	-32.1118
8	3	0.00e+00	-2.64977	-37.0879
8	6	0.00e+00	-4.32105	-41.8396
10	0	0.00e+00	-0.86876	-39.838
10	3	0.00e+00	-2.23819	-44.9222
10	6	0.00e+00	-3.70594	-49.8064

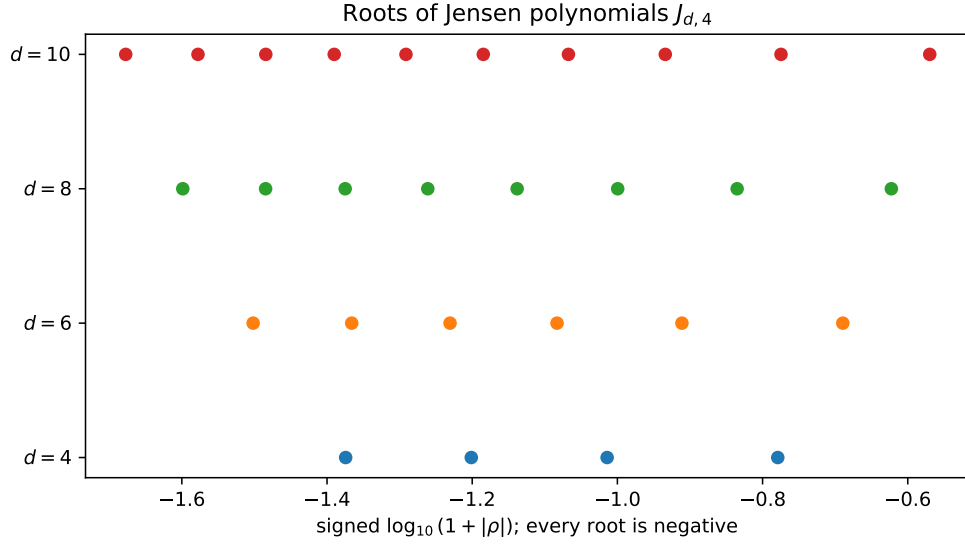


Figure 1: Selected shifted Jensen roots. The horizontal coordinate is compressed logarithmically; all roots are negative, in agreement with [Theorem 3.5](#).

4 Bernoulli, Bell, and symmetric-function formulas

4.1 Spectral power sums

Proposition 4.1 (Valuation Dirichlet series). *For $\Re s > 1$,*

$$\sum_{m=1}^{\infty} \frac{d_b(m)}{m^s} = \frac{\zeta(s)}{1 - b^{-s}}. \quad (19)$$

Consequently, if

$$p_r^{(b)} := \sum_{m=1}^{\infty} \frac{d_b(m)}{\pi^{2r} m^{2r}}, \quad (20)$$

then

$$p_r^{(b)} = \frac{\zeta(2r)}{\pi^{2r}(1 - b^{-2r})}. \quad (21)$$

For $b = 2$,

$$p_r^{(2)} = \frac{(-1)^{r+1} 2^{4r-1} B_{2r}}{(2r)!(2^{2r} - 1)}. \quad (22)$$

Proof. Since $\nu_b(m) = \sum_{j \geq 1} \mathbf{1}_{bj|m}$, absolute convergence gives

$$\sum_{m \geq 1} \frac{\nu_b(m)}{m^s} = \sum_{j \geq 1} \sum_{n \geq 1} \frac{1}{(bjn)^s} = \zeta(s) \frac{b^{-s}}{1 - b^{-s}}.$$

Adding $\zeta(s)$ proves (19); Euler's evaluation of $\zeta(2r)$ gives (22). $\square$

4.2 Newton recurrence and Bell polynomial

Taking logarithms in (11) gives, for $|z| < \pi^2$,

$$\log A_b(z) = \sum_{r=1}^{\infty} \frac{(-1)^{r-1}}{r} p_r^{(b)} z^r. \quad (23)$$

Differentiation yields the exact Newton recurrence

$$na_n^{(b)} = \sum_{r=1}^n (-1)^{r-1} p_r^{(b)} a_{n-r}^{(b)}, \quad a_0^{(b)} = 1. \quad (24)$$

Equivalently, with B_n the complete exponential Bell polynomial,

$$a_n^{(b)} = \frac{1}{n!} B_n(p_1^{(b)}, -1!p_2^{(b)}, 2!p_3^{(b)}, \dots, (-1)^{n-1}(n-1)!p_n^{(b)}). \quad (25)$$

The elementary-symmetric determinant is

$$a_n^{(b)} = \frac{1}{n!} \det \begin{pmatrix} p_1 & 1 & 0 & \cdots & 0 \\ p_2 & p_1 & 2 & \cdots & 0 \\ p_3 & p_2 & p_1 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & n-1 \\ p_n & p_{n-1} & \cdots & p_2 & p_1 \end{pmatrix}, \quad (26)$$

where $p_r = p_r^{(b)}$.

Remark 4.2. The repository already contains extensive Bell–Bernoulli moment algebra. The new feature here is the interpretation of exactly the same coefficients as elementary symmetric functions of the arithmetic spectrum

$$\left\{ \frac{1}{\pi^2 m^2} \text{ repeated } d_b(m) \text{ times} \right\},$$

which is what makes total positivity and Jensen hyperbolicity automatic.

Numerical experiment 4.3 (Exact dyadic moments). The first values generated by (24) are:

n	$a_n = 4^n \mu_{2n} / (2n)!$	μ_{2n}	decimal μ_{2n}
0	1	1	1.000000000000
1	$\frac{2}{9}$	$\frac{1}{9}$	0.111111111111
2	$\frac{38}{2025}$	$\frac{19}{675}$	0.0281481481481
3	$\frac{2332}{2679075}$	$\frac{583}{59535}$	0.00979255899891
4	$\frac{265618}{10247461875}$	$\frac{1333}{132809}$	0.00408245822334
5	$\frac{187362796}{345944065438125}$	$\frac{46840699}{24405225075}$	0.00191928977733
6	$\frac{16275962240644}{1933714893977851359375}$	$\frac{4068990560161}{4133856862760625}$	0.000984308527181
7	$\frac{9636538427610488}{95040153324117416461921875}$	$\frac{1204567303451311}{2232691548877164375}$	0.000539513532022

The values $\mu_2 = 1/9$, $\mu_4 = 19/675$, and $\mu_6 = 583/59535$ agree with the independent probabilistic recurrences in the existing corpus.

5 Non-holonomicity and non-P-recursive moments

5.1 A zero-multiplicity obstruction

Lemma 5.1 (Multiplicity bound for D-finite entire functions). *Let $f \not\equiv 0$ be an entire solution of*

$$P_r(z)f^{(r)}(z) + P_{r-1}(z)f^{(r-1)}(z) + \cdots + P_0(z)f(z) = 0, \quad (27)$$

where $P_r \not\equiv 0$ is a polynomial. At every point z_0 with $P_r(z_0) \neq 0$, the multiplicity of a zero of f is at most $r - 1$.

Proof. If f vanished to order at least r at an ordinary point z_0 , then $f(z_0) = \cdots = f^{(r-1)}(z_0) = 0$. Equation (27) gives $f^{(r)}(z_0) = 0$. Differentiating the equation repeatedly then forces every derivative of f at z_0 to vanish, contradicting $f \not\equiv 0$. $\square$

Theorem 5.2 (Non-D-finiteness). *For every integer $b \geq 2$, neither Φ_b nor A_b is D-finite over $\mathbb{C}(z)$.*

Proof. At $m = b^k$, Theorem 2.1 gives

$$\text{ord}_{z=m} \Phi_b = 1 + \nu_b(b^k) = k + 1.$$

The multiplicities are unbounded. If Φ_b satisfied an equation of order r , all but finitely many of these zeros would be ordinary points of that equation, contradicting Theorem 5.1. The zeros of A_b at $-\pi^2 b^{2k}$ have the same unbounded multiplicities, so the identical argument applies. $\square$

Corollary 5.3 (The moments are not P-recursive). *The coefficient sequence $(a_n^{(b)})_{n \geq 0}$ is not P-recursive. In particular, the Rvachev even moments $(\mu_{2n})_{n \geq 0}$ are not P-recursive.*

Proof. A power series is D-finite if and only if its coefficient sequence is P-recursive [15]. Thus the first assertion follows from Theorem 5.2. In the dyadic case $a_n = 4^n \mu_{2n} / (2n)!$. Multiplication or division by this nonzero hypergeometric factor converts a polynomial-coefficient recurrence into another one after denominators are cleared. Hence P-recursiveness of one sequence would imply that of the other. $\square$

Corollary 5.4 (Non-D-finite Thorin transforms). *For every $t > 0$, the germ $A_b(z)^{-t}$ is not D-finite.*

Proof. For integral t , the continuation has poles at every $-\pi^2 m^2$; for nonintegral t , it has branch singularities there. A D-finite germ has only finitely many finite singular points, all among the roots of a leading polynomial. The infinite singular set is therefore impossible. $\square$

6 Cartwright class and digital zero counts

Theorem 6.1 (Cartwright classification). *For every integer $b \geq 2$, Φ_b is a Cartwright-class entire function of exact exponential type*

$$\tau_b = \frac{\pi b}{b-1}. \quad (28)$$

Proof. The random variable X_b in (5) has support $[-R_b, R_b]$ with $R_b = b/[2(b-1)]$. Its Fourier–Laplace transform therefore has exponential type at most $2\pi R_b = \pi b/(b-1)$. The support reaches arbitrarily close to both endpoints, so the type is exact; equivalently, one

can evaluate $\Phi_b(iy) = \mathbb{E}e^{2\pi y X_b}$ and use a fixed positive mass in $(R_b - \varepsilon, R_b]$. On the real axis every sinc factor has modulus at most one, hence $|\Phi_b(x)| \leq 1$ and

$$\int_{\mathbb{R}} \frac{\log^+ |\Phi_b(x)|}{1+x^2} dx = 0.$$

This is the Cartwright condition; standard entire-function background is in [11]. $\square$

Let $s_b(N)$ denote the sum of the base- b digits of N .

Proposition 6.2 (Exact digital zero count). *The total multiplicity of the positive zeros $1, 2, \dots, N$ is*

$$M_b(N) := \sum_{m=1}^N d_b(m) = \frac{bN - s_b(N)}{b-1}. \quad (29)$$

Consequently, the number of nonzero zeros in $|z| \leq R$, counted with multiplicity, is

$$n_b(R) = \frac{2(b\lfloor R \rfloor - s_b(\lfloor R \rfloor))}{b-1}. \quad (30)$$

In particular,

$$n_b(R) = \frac{2b}{b-1}R + O(\log R), \quad (31)$$

whose leading density equals $2\tau_b/\pi$, as predicted by Cartwright theory.

Proof. Using $\nu_b(m) = \sum_{j \geq 1} \mathbf{1}_{b^j | m}$,

$$M_b(N) = N + \sum_{j \geq 1} \left\lfloor \frac{N}{b^j} \right\rfloor.$$

The standard base- b digit identity $\sum_{j \geq 1} \lfloor N/b^j \rfloor = (N - s_b(N))/(b-1)$ gives (29); reflection gives (30). $\square$

Remark 6.3 (Thue–Morse consistency). For $b = 2$, $M_2(N) = 2N - s_2(N)$, so

$$(-1)^{M_2(N)} = (-1)^{s_2(N)}.$$

Since $\Phi(0) > 0$, the sign of Φ on the lobe $(N, N+1)$ is therefore the Thue–Morse sign. This recovers, but does not duplicate as a novelty claim, the repository’s formalized lobe-sign theorem.

7 Complete Bernstein structure and a Thorin gamma dual

7.1 Stieltjes logarithmic derivative

Theorem 7.1 (Stieltjes and complete Bernstein structure). *For $z > 0$,*

$$\frac{A'_b(z)}{A_b(z)} = \sum_{m=1}^{\infty} \frac{d_b(m)}{z + \pi^2 m^2}. \quad (32)$$

Thus A'_b/A_b is a Stieltjes function and $\log A_b$ is a complete Bernstein function on $(0, \infty)$.

Proof. Logarithmic differentiation of (11) is justified locally uniformly away from the negative zeros and gives (32). It is the Stieltjes transform of the positive discrete measure

$$\rho_b := \sum_{m \geq 1} d_b(m) \delta_{\pi^2 m^2}. \quad (33)$$

A nonnegative function vanishing at zero whose derivative is Stieltjes is a complete Bernstein function; alternatively, sum the elementary complete Bernstein functions $d_b(m) \log(1 + z/(\pi^2 m^2))$; see [14]. $\square$

7.2 Generalized gamma convolution

Theorem 7.2 (Thorin spectral law). *For every $t > 0$, $A_b(z)^{-t}$ is the Laplace transform of the almost surely finite random series*

$$Y_{b,t} := \sum_{m=1}^{\infty} G_{m,t}, \quad G_{m,t} \sim \text{Gamma}(td_b(m), \text{rate } \pi^2 m^2), \quad (34)$$

with independent summands. Hence $Y_{b,t}$ is a generalized gamma convolution with Thorin measure

$$U_{b,t} = t \sum_{m=1}^{\infty} d_b(m) \delta_{\pi^2 m^2}. \quad (35)$$

Proof. The Laplace transform of a gamma variable of shape α and rate λ is $(1 + z/\lambda)^{-\alpha}$. Independence and (11) give

$$\mathbb{E}e^{-zY_{b,t}} = \prod_{m \geq 1} \left(1 + \frac{z}{\pi^2 m^2}\right)^{-td_b(m)} = A_b(z)^{-t}.$$

Moreover,

$$\mathbb{E}Y_{b,t} = t \sum_{m \geq 1} \frac{d_b(m)}{\pi^2 m^2} = t p_1^{(b)} < \infty,$$

so the nonnegative series is finite almost surely. The product is exactly the canonical Thorin representation in the theory of generalized gamma convolutions [5, 10]. $\square$

Corollary 7.3 (Cumulants). *The cumulants of $Y_{b,t}$ are*

$$\kappa_r(Y_{b,t}) = t(r-1)!p_r^{(b)}. \quad (36)$$

For $b = 2$,

$$\mathbb{E}Y_{2,t} = \frac{2t}{9}, \quad \text{Var}(Y_{2,t}) = \frac{8t}{675}. \quad (37)$$

Proof. Use (23) in $\log \mathbb{E}e^{sY_{b,t}} = -t \log A_b(-s)$ and compare with the cumulant series. $\square$

Corollary 7.4 (Scale self-decomposition). *Let*

$$S(z) := \frac{\sinh \sqrt{z}}{\sqrt{z}}. \quad (38)$$

Then

$$A_b(z) = S(z)A_b(z/b^2). \quad (39)$$

Consequently,

$$Y_{b,t} \stackrel{d}{=} Z_t + b^{-2}Y'_{b,t}, \quad (40)$$

where $Y'_{b,t}$ is an independent copy and

$$\mathbb{E}e^{-zZ_t} = S(z)^{-t} = \prod_{n \geq 1} \left(1 + \frac{z}{\pi^2 n^2}\right)^{-t}. \quad (41)$$

Proof. Separate the $j = 0$ factor in (10); then interpret the factors as Laplace transforms. $\square$

8 The arithmetic heat trace

8.1 Definition, q-renewal, and Malmsten representation

Define

$$\Theta_b(x) := \sum_{m=1}^{\infty} d_b(m) e^{-\pi^2 m^2 x}, \quad x > 0, \quad (42)$$

and the ordinary positive theta tail

$$\vartheta_+(x) := \sum_{n=1}^{\infty} e^{-\pi^2 n^2 x}. \quad (43)$$

Proposition 8.1 (Exact theta renewal). *For every $x > 0$,*

$$\Theta_b(x) = \sum_{j=0}^{\infty} \vartheta_+(b^{2j}x), \quad (44)$$

and hence

$$\Theta_b(x) - \Theta_b(b^2x) = \vartheta_+(x). \quad (45)$$

Proof. Use $d_b(m) = \sum_{j \geq 0} \mathbf{1}_{b^j | m}$, then set $m = b^j n$. All terms are nonnegative, so Tonelli's theorem applies. $\square$

Proposition 8.2 (Malmsten–Lévy integral). *For $z \geq 0$,*

$$\frac{A'_b(z)}{A_b(z)} = \int_0^{\infty} e^{-zx} \Theta_b(x) \, dx, \quad (46)$$

$$\log A_b(z) = \int_0^{\infty} (1 - e^{-zx}) \frac{\Theta_b(x)}{x} \, dx. \quad (47)$$

Thus the Lévy density of $Y_{b,t}$ is

$$\ell_{b,t}(x) = \frac{t}{x} \Theta_b(x). \quad (48)$$

Proof. Insert $(z + \lambda)^{-1} = \int_0^{\infty} e^{-(z+\lambda)x} \, dx$ in (32). Integrating from 0 to z , or using $\log(1 + z/\lambda) = \int_0^{\infty} (1 - e^{-zx}) e^{-\lambda x} \, dx/x$, gives (47). $\square$

8.2 Mellin transform and log-periodic skeleton

Theorem 8.3 (Mellin transform). *For $\Re s > 1/2$,*

$$\int_0^{\infty} x^{s-1} \Theta_b(x) \, dx = \Gamma(s) \pi^{-2s} \frac{\zeta(2s)}{1 - b^{-2s}}. \quad (49)$$

Proof. Termwise Mellin transformation of (42) gives

$$\Gamma(s) \pi^{-2s} \sum_{m \geq 1} \frac{d_b(m)}{m^{2s}},$$

and Theorem 4.1 completes the proof. $\square$

Put $L = \log b$ and

$$s_k := \frac{\pi i k}{L}, \quad k \in \mathbb{Z} \setminus \{0\}. \quad (50)$$

Define

$$c_{b,k} := \frac{\Gamma(s_k) \pi^{-2s_k} \zeta(2s_k)}{2L}, \quad (51)$$

and the absolutely convergent one-periodic Fourier series

$$P_b(u) := 2\Re \sum_{k=1}^{\infty} c_{b,k} e^{2\pi i k u}. \quad (52)$$

Absolute convergence follows from the exponential decay of $\Gamma(iy)$.

Theorem 8.4 (Complete algebraic/log-periodic skeleton). *As $x \downarrow 0$, for every $N > 0$,*

$$\Theta_b(x) = \mathcal{H}_b(x) + O_N(x^N), \quad (53)$$

where

$$\mathcal{H}_b(x) = \frac{b}{2(b-1)\sqrt{\pi x}} - \frac{\log(1/x)}{4 \log b} - \frac{1}{4} + \frac{\gamma - 2 \log 2}{4 \log b} + P_b\left(\frac{\log(1/x)}{2 \log b}\right). \quad (54)$$

Proof. Mellin inversion of (49) starts on a vertical line; the harmonic-sum methodology is standard [8]. $\Re s > 1/2$. The integrand has:

- a simple pole at $s = 1/2$, with residue $b/[2(b-1)\sqrt{\pi}]$;
- a double pole at $s = 0$;
- simple poles at $s = s_k$, $k \neq 0$, from $1 - b^{-2s} = 0$.

The poles of $\Gamma(s)$ at negative integers are canceled by the trivial zeros of $\zeta(2s)$. Near zero,

$$\Gamma(s) \pi^{-2s} \frac{\zeta(2s)}{1 - b^{-2s}} = -\frac{1}{4L} \frac{1}{s^2} + \left[-\frac{1}{4} + \frac{\gamma - 2 \log 2}{4L} \right] \frac{1}{s} + O(1). \quad (55)$$

Multiplication by x^{-s} supplies the logarithmic term in (54). The nonzero imaginary poles supply (52). Shifting the contour arbitrarily far left gives $O_N(x^N)$, since no further poles remain and $\Gamma(s)$ decays exponentially on vertical lines. $\square$

Numerical experiment 8.5 (Periodic coefficients).

b	k	$ c_{b,k} $	$\arg c_{b,k}$
2	1	0.00111534751852	-2.407777457
2	2	1.22755013029e-6	1.762155882
2	3	1.06068491216e-9	2.991879099
3	1	0.00630577173822	-0.4450285399
3	2	0.000101004691651	2.396816114
3	3	1.5958886918e-6	2.427407365
4	1	0.0105852315462	0.2218422212
4	2	0.000557673759259	-2.407777457
4	3	5.3859323085e-6	0.5476719675

For $b = 2$, the first harmonic has amplitude about $2|c_{2,1}| = 2.23 \times 10^{-3}$, while the second is already of order 10^{-6} . This explains the nearly sinusoidal graph in Fig. 2.

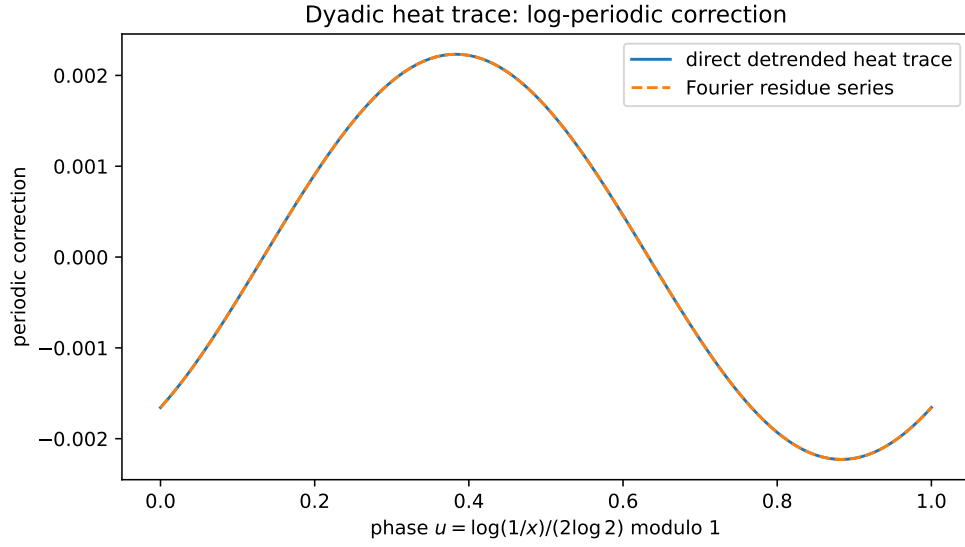


Figure 2: Directly summed dyadic heat trace after removal of the elementary terms, compared with the Fourier series P_2 .

8.3 Exact beyond-all-orders remainder

The all-orders estimate in [Theorem 8.4](#) conceals a simpler exact identity.

Theorem 8.6 (Jacobi-dual transseries). *For every $x > 0$,*

$$\Theta_b(x) = \mathcal{H}_b(x) - \frac{1}{\sqrt{\pi x}} \sum_{j=1}^{\infty} b^j \sum_{n=1}^{\infty} \exp\left(-\frac{b^{2j} n^2}{x}\right). \quad (56)$$

In particular, $\Theta_b(x) < \mathcal{H}_b(x)$ for every $x > 0$, and

$$\mathcal{H}_b(x) - \Theta_b(x) \sim \frac{b}{\sqrt{\pi x}} e^{-b^2/x} \quad (x \downarrow 0). \quad (57)$$

More precisely, the relative error in (57) is $O(e^{-c_b/x})$, where

$$c_b = \min\{3b^2, b^4 - b^2\}. \quad (58)$$

Proof. Jacobi inversion gives

$$\vartheta_+(x) = \frac{1}{2\sqrt{\pi x}} - \frac{1}{2} + \frac{1}{\sqrt{\pi x}} \sum_{n=1}^{\infty} e^{-n^2/x}. \quad (59)$$

A direct calculation from (54), using the one-periodicity of P_b , shows

$$\mathcal{H}_b(x) - \mathcal{H}_b(b^2 x) = \frac{1}{2\sqrt{\pi x}} - \frac{1}{2}. \quad (60)$$

Let $R_b = \Theta_b - \mathcal{H}_b$. Combining this identity with (45) and (59) gives

$$R_b(x) - R_b(b^2 x) = \frac{1}{\sqrt{\pi x}} \sum_{n \geq 1} e^{-n^2/x}. \quad (61)$$

Replace x by x/b^2 and iterate toward zero:

$$R_b(x) = R_b(x/b^{2J}) - \frac{1}{\sqrt{\pi x}} \sum_{j=1}^J b^j \sum_{n \geq 1} e^{-b^{2j} n^2 / x}.$$

By [Theorem 8.4](#), $R_b(y) \rightarrow 0$ faster than every power as $y \downarrow 0$. Letting $J \rightarrow \infty$ proves (56). The leading term is $(j, n) = (1, 1)$; the next exponential action is $\min(4b^2, b^4)$, giving the stated relative error. $\square$

Corollary 8.7 (Lower-Lambert inversion of the heat defect). *Let*

$$\Delta_b(x) := \mathcal{H}_b(x) - \Theta_b(x). \quad (62)$$

For any asymptotic inverse branch as $r \downarrow 0$,

$$\Delta_b^{-1}(r) \sim \frac{2b^2}{-W_{-1}(-2\pi r^2)}. \quad (63)$$

Proof. Square (57) and put $y = 2b^2/x$:

$$r^2 \sim \frac{b^2}{\pi x} e^{-2b^2/x} = \frac{y}{2\pi} e^{-y}.$$

The relevant large positive solution is $y \sim -W_{-1}(-2\pi r^2)$, using the lower real branch described in [6]. $\square$

Remark 8.8. The same lower Lambert branch governs the inverse-Fabius endpoint theory, but it arises here from a different mechanism: Jacobi-dual exponential action rather than the original Fabius saddle. This provides a concrete route for transferring the repository’s Lambert phase and phase-locked extrapolation machinery to spectral heat defects.

Numerical experiment 8.9 (Exact closure). The table compares direct summation, the skeleton, and the exact dual remainder.

b	x	direct Θ_b	skeleton $\mathcal{H}_b$	dual remainder	closure error
2	10^{-2}	3.441187881196	3.441187881196	-2.161e-173	1.05e-68
2	10^{-4}	52.55506546033	52.55506546033	-1.876e-17370	1.0e-65
2	10^{-6}	558.6629413948	558.6629413948	-1.333e-1737175	2.46e-64
3	10^{-2}	2.761629350382	2.761629350382	-2.309e-390	2.62e-69
3	10^{-4}	39.79336411784	39.79336411784	-5.311e-39085	6.24e-66
3	10^{-6}	419.5670190346	419.5670190346	-7.788e-3908648	1.74e-64

The transformed remainder in [Fig. 3](#) approaches b^2 , as forced by (57).

9 Exact modular duality for the infinite sinc moment product

9.1 Laplace transform of the heat identity

Set

$$B_b := -\frac{1}{4} + \frac{\gamma - 2 \log 2}{4 \log b}. \quad (64)$$

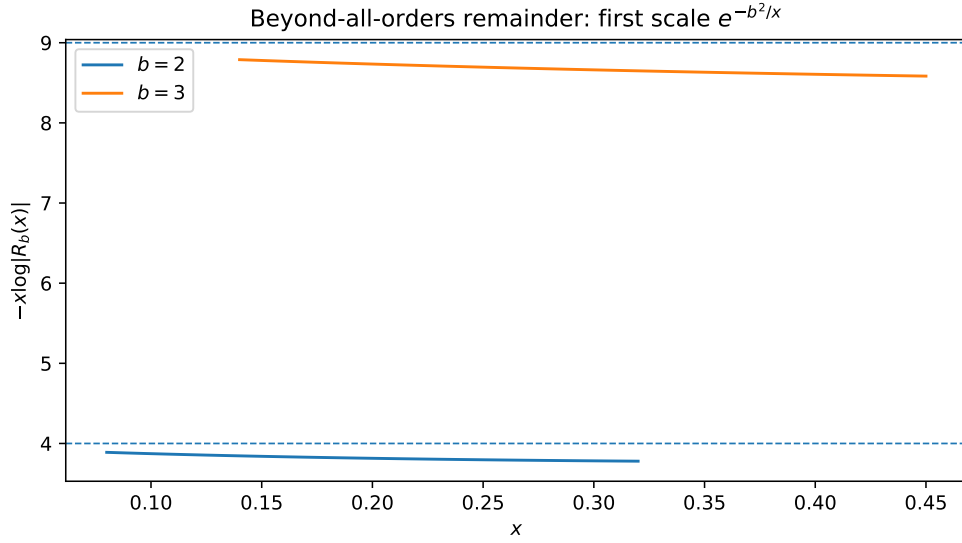


Figure 3: The exact remainder has first action b^2/x . The dashed levels are the proven limits 4 and 9.

The Laplace transform of (56) can be evaluated term by term. The elementary integrals are

$$\int_0^\infty e^{-zx} x^{-1/2} dx = \sqrt{\pi} z^{-1/2}, \quad (65)$$

$$\int_0^\infty e^{-zx} \log(1/x) dx = \frac{\log z + \gamma}{z}, \quad (66)$$

$$\int_0^\infty x^{-1/2} e^{-zx-a^2/x} dx = \sqrt{\frac{\pi}{z}} e^{-2a\sqrt{z}}. \quad (67)$$

For s_k from (50), define

$$q_{b,k} := -c_{b,k} \Gamma(-s_k) \quad (68)$$

and

$$Q_b(v) := 2\Re \sum_{k=1}^\infty q_{b,k} e^{2\pi i k v}. \quad (69)$$

This Fourier series is absolutely convergent.

Theorem 9.1 (Exact modular-dual formula). *There exists a real constant C_b such that, for every $z > 0$,*

$$\begin{aligned} \log A_b(z) = & \frac{b}{b-1} \sqrt{z} - \frac{(\log z)^2}{8 \log b} - \left(\frac{1}{4} + \frac{\log 2}{2 \log b} \right) \log z \\ & + C_b + Q_b\left(\frac{\log z}{2 \log b}\right) - \sum_{j=1}^\infty \log(1 - e^{-2b^j \sqrt{z}}). \end{aligned} \quad (70)$$

One rapidly convergent definition of the constant is

$$C_b = \log A_b(1) - \frac{b}{b-1} - Q_b(0) + \sum_{j=1}^\infty \log(1 - e^{-2b^j}). \quad (71)$$

Proof. By (46), Laplace transformation of the skeleton in (54) gives

$$\frac{d}{dz} \log A_b(z) = \frac{b}{2(b-1)} z^{-1/2} - \frac{\log z + \gamma}{4(\log b)z} + \frac{B_b}{z}$$

$$+ 2\Re \sum_{k \geq 1} c_{b,k} \Gamma(1 - s_k) z^{s_k - 1} \\ - z^{-1/2} \sum_{j \geq 1} \frac{b^j}{e^{2bj\sqrt{z}} - 1}.$$

Integrating in z gives the first line of (70); the logarithmic coefficient simplifies because

$$B_b - \frac{\gamma}{4 \log b} = -\frac{1}{4} - \frac{\log 2}{2 \log b}.$$

Furthermore, $\Gamma(1 - s_k)/s_k = -\Gamma(-s_k)$, producing (68). Finally,

$$\frac{d}{dz} \left[-\log(1 - e^{-2a\sqrt{z}}) \right] = -\frac{a}{\sqrt{z}(e^{2a\sqrt{z}} - 1)}.$$

Thus the last derivative integrates to the lacunary logarithmic product. The remaining integration constant is fixed at $z = 1$, giving (71). $\square$

9.2 A partition product hidden in the sinc product

Let $q = e^{-2\sqrt{z}}$. The final factor in (70) exponentiates to

$$\prod_{j=1}^{\infty} (1 - q^{b^j})^{-1}. \quad (72)$$

This is the ordinary generating function for partitions into parts from $\{b, b^2, b^3, \dots\}$. In the dyadic case it is the binary-partition product with the part 1 omitted. Hence the Rvachev moment product has an exact modular dual in which an infinite sinc convolution is paired with a lacunary partition product.

Corollary 9.2 (Large- z transseries). *Uniformly for $z \rightarrow +\infty$,*

$$\log A_b(z) = \frac{b}{b-1} \sqrt{z} - \frac{(\log z)^2}{8 \log b} - \left(\frac{1}{4} + \frac{\log 2}{2 \log b} \right) \log z + C_b \quad (73)$$

$$+ Q_b \left(\frac{\log z}{2 \log b} \right) + e^{-2b\sqrt{z}} + O(e^{-4b\sqrt{z}} + e^{-2b^2\sqrt{z}}). \quad (74)$$

Proof. Expand the last logarithm in (70) as $-\log(1 - w) = w + O(w^2)$. The first action is $j = 1$, with $w = e^{-2b\sqrt{z}}$. $\square$

Numerical experiment 9.3 (Modular-dual identity).

b	z	direct $\log A_b(z)$	modular-dual formula	absolute error
2	1	0.21663004400518	0.21663004400518	2.07e-76
2	10	1.8392973325664	1.8392973325664	7.91e-67
2	100	10.919419442711	10.919419442711	1.19e-66
2	1000	47.65739566566	47.65739566566	4.95e-68
3	1	0.18220373098747	0.18220373098747	3.45e-77
3	10	1.5178809593783	1.5178809593783	2.99e-68
3	100	8.6627244525758	8.6627244525758	6.32e-68
3	1000	36.778663212226	36.778663212226	1.0e-67

The constants used in the numerical implementation are

b	C_b
2	-1.802192545994131991608
3	-1.320073991706705495258
4	-1.161026465707045412821
5	-1.082717524532689081388

Agreement is limited only by the requested working precision.

10 Consequences for computation and approximation

10.1 Certified Stieltjes–Padé control

Because A'_b/A_b is a Stieltjes function, its diagonal and subdiagonal Padé approximants have negative real poles and positive residues, and they form the classical alternating upper/lower bounds on the positive axis. This yields a new certified route to the saddle equations that occur in the Fabius endpoint theory:

1. compute the exact rational coefficients $p_r^{(b)}$ from Bernoulli numbers;
2. form Stieltjes Padé approximants to A'_b/A_b ;
3. integrate the rational bounds to bracket $\log A_b$;
4. transfer the bounds through the monotone saddle map and, when needed, through the lower Lambert branch.

Unlike an unconstrained power-series Padé computation, every approximant respects the correct negative spectral axis.

10.2 Moment cones and Legendre expansions

Every even Legendre coefficient of up is a finite rational linear combination of $\mu_0, \mu_2, \dots, \mu_{2n}$. The PF_∞ constraints and the Jensen hyperbolicity hierarchy therefore place the admissible moment vectors in a nested family of explicit determinantal cones. This does not by itself determine the sign of each Legendre coefficient, but it supplies:

- rigorous conditioning inequalities for the moment-to-Legendre triangular transform;
- determinant certificates that can accompany the repository's exact Legendre-energy calculations;
- hyperbolic-polynomial barriers for moment truncation and rational reconstruction.

A promising next step is to combine these cones with the existing shifted-up synthesis of Legendre polynomials.

10.3 Inverse-Fabius transfer

The new formulas enter inverse-Fabius analysis in three ways.

1. [Theorem 8.7](#) gives a second, independent lower- W inversion problem with the same logarithmic anatomy as the endpoint inverse of F .
2. The exact periodic functions P_b and Q_b have period $2 \log b$ in the raw logarithmic variable, reflecting the quadratic spectral lift $m \mapsto \pi^2 m^2$. They can be treated by the repository's phase-locked Richardson filters after a period rescaling.

3. The Stieltjes bounds above can replace floating-point evaluation of the sinc product in saddle iterations by certified rational inequalities.

11 Conjectures and frontier problems

Conjecture 11.1 (Differential transcendence). *For every integer $b \geq 2$, the functions Φ_b and A_b are differentially transcendental over $\mathbb{C}(z)$.*

Evidence. [Theorem 5.2](#) excludes linear differential equations. The exact divisor has unbounded arithmetic multiplicities, while the scale relation $A_b(z) = S(z)A_b(z/b^2)$ couples differential and Mahler-type structures. Non-D-finiteness alone is not enough—for example, some familiar meromorphic functions are differentially algebraic—so a proof will require a genuinely nonlinear obstruction, perhaps through Nevanlinna theory for differential equations with a prescribed divisor.

Conjecture 11.2 (Strict total positivity after forced zeros). *Every Toeplitz minor of $(a_n^{(b)})$ that is not forced to vanish by the zero convention $a_n = 0$ for $n < 0$ is strictly positive.*

Finite truncations have strictly positive spectral parameters, but repeated zeros in the generating polynomial make a direct limiting argument delicate. A proof would strengthen variation diminution to strict sign-regularity.

Conjecture 11.3 (Hermite universality of Jensen polynomials). *For each fixed degree d , an affine normalization of $J_{d,n}^{(2)}$ converges coefficientwise to the degree- d Hermite polynomial as $n \rightarrow \infty$.*

The repository’s all-order endpoint asymptotics should give the required regular expansion of $\log \gamma_n$. Establishing the Griffin–Ono–Rolen–Zagier hypotheses [9] would convert the exact hyperbolicity theorem into a universal limiting shape theorem.

Conjecture 11.4 (Closed form for C_b). *The constant C_b in (70) can be expressed using a base- b Glaisher–Kinkelin/Barnes-type constant and elementary combinations of $\log b$, $\log(2\pi)$, and zeta derivatives at zero.*

The double Mellin pole explains the quadratic logarithm, while C_b is the finite part left by the same spectral determinant. A zeta-regularized determinant of the spectral multiset

$$\{\pi^2 m^2 : m \geq 1, \text{ with multiplicity } d_b(m)\}$$

is the natural candidate.

Problem 11.5 (Nonintegral geometric bases). *For real $\beta > 1$, the same summable real-zero product and compact-support argument already show that*

$$\Phi_\beta(z) = \prod_{j \geq 0} \operatorname{sinc}(\pi z / \beta^j)$$

belongs to the Laguerre–Pólya and Cartwright classes. Classify instead its D-finiteness, zero-collision multiplicities, exact counting fluctuations, and heat-trace periodicity. The zero set is the union $\{\pm n\beta^j\}$, and collisions depend on multiplicative relations among powers of β . The integer-base proof by unbounded multiplicity no longer applies generically.

Problem 11.6 (Orthogonal-polynomial asymptotics). *Use the known endpoint flatness of up to place it precisely in the Szegő class and determine the first correction to the Jacobi recurrence coefficients. The log-squared endpoint profile suggests a slowly varying, non-Fisher–Hartwig correction rather than a pure power law.*

Problem 11.7 (Lean formalization). *A realistic formalization sequence is:*

1. derive (7) from the existing sinc canonical product and weighted-scale multiplicity API;
2. prove the D -finite multiplicity lemma in an abstract analytic ODE setting;
3. formalize (19) and the exact digital count;
4. define Θ_b , prove (44), and import a Jacobi theta transformation;
5. prove the exact remainder (56);
6. formalize the Stieltjes/Laplace identities and the gamma-series probability construction.

The LP/PF/Jensen portion will require more library infrastructure than the heat-trace and non- D -finite portions.

12 Summary of the new theorem network

The report’s deductions can be read as two interacting chains.

Entire-function chain:

$$\begin{aligned} \text{integer divisor} &\implies \mathcal{LP}/\mathcal{LP}\text{-I} \implies \text{PF}_\infty \text{ moments} \\ &\implies \text{Jensen hyperbolicity} \implies \text{Turán inequalities.} \end{aligned}$$

while the same unbounded multiplicities imply

$$\text{integer divisor} \implies \text{non-}D\text{-finite} \implies \text{non-P-recursive moments.}$$

Spectral-probability chain:

$$\begin{aligned} \frac{A'_b}{A_b} \text{ Stieltjes} &\implies \log A_b \text{ complete Bernstein} \\ &\implies A_b^{-t} \text{ GGC} \implies \Theta_b \text{ Lévy heat trace.} \end{aligned}$$

Then

$$\begin{aligned} \Theta_b \text{ Mellin poles} &\implies \text{log-periodic skeleton} \\ \xRightarrow{\text{Jacobi}} \text{exact flat remainder} &\xRightarrow{\text{Laplace}} \text{lacunary partition duality.} \end{aligned}$$

These results are structurally compatible with the repository’s q -series, Bell–Bernoulli, Thue–Morse, Lambert– W , inverse-Fabius, and Legendre programs, but they are not restatements of the current documents.

A Reproducibility

The archive contains `numerical_experiments.py`. From the report directory, run

```
python numerical_experiments.py --precision 75 --moments 28
latexmk -pdf -interaction=nonstopmode fabius_frontier_report.tex
```

The program writes exact tables to `generated/` and figures to `figures/`. The exact arithmetic uses `sympy.Rational`; no decimal reconstruction is used for the moment table. Heat sums and periodic coefficients use arbitrary-precision `mpmath`. The modular-dual and Jacobi identities are checked by evaluating both independently computed sides.

B Selected numerical constants

The first dyadic periodic coefficients are

$$\begin{aligned} |c_{2,1}| &\approx 1.11534751852 \times 10^{-3}, & |c_{2,2}| &\approx 1.22755013029 \times 10^{-6}, \\ |c_{2,3}| &\approx 1.06068491216 \times 10^{-9}. \end{aligned}$$

The modular-dual constants begin

$$C_2 \approx -1.80219254599413199161, \quad C_3 \approx -1.32007399170670549526.$$

These decimal values are diagnostics, not definitions; (71) is the mathematical definition used in the report.

References

- [1] M. Aissen, A. Edrei, I. J. Schoenberg, and A. Whitney, On the generating functions of totally positive sequences, *Proc. Natl. Acad. Sci. USA* **37** (1951), 303–307.
- [2] M. Aissen, A. Edrei, I. J. Schoenberg, and A. Whitney, On the generating functions of totally positive sequences I, *J. Analyse Math.* **2** (1952), 93–103.
- [3] J. Arias de Reyna, An infinitely differentiable function with compact support: definition and properties, arXiv:1702.05442 (English translation of the 1982 article).
- [4] J. Arias de Reyna, Arithmetic of the Fabius function, *Integers* **18** (2018), A51; arXiv:1702.06487.
- [5] L. Bondesson, *Generalized Gamma Convolutions and Related Classes of Distributions and Densities*, Lecture Notes in Statistics 76, Springer, 1992.
- [6] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey, and D. E. Knuth, On the Lambert W function, *Adv. Comput. Math.* **5** (1996), 329–359.
- [7] J. Fabius, A probabilistic example of a nowhere analytic C^∞ -function, *Z. Wahrscheinlichkeitstheorie verw. Gebiete* **5** (1966), 173–174.
- [8] P. Flajolet, X. Gourdon, and P. Dumas, Mellin transforms and asymptotics: harmonic sums, *Theoret. Comput. Sci.* **144** (1995), 3–58.
- [9] M. J. Griffin, K. Ono, L. Rolén, and D. Zagier, Jensen polynomials for the Riemann zeta function and other sequences, *Proc. Natl. Acad. Sci. USA* **116** (2019), 11103–11110.
- [10] L. F. James, B. Roynette, and M. Yor, Generalized gamma convolutions, Dirichlet means, Thorin measures, with explicit examples, *Probability Surveys* **5** (2008), 346–415.
- [11] B. Ya. Levin, *Lectures on Entire Functions*, Translations of Mathematical Monographs 150, American Mathematical Society, 1996.
- [12] V. Reshetnikov, *ProveIt: Analysis/FabiusFunction*, Lean 4 formal development and documentation corpus, <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction>.
- [13] V. A. Rvachev, Compactly supported solutions of functional-differential equations and their applications, *Russian Math. Surveys* **45** (1990), no. 1, 87–120.
- [14] R. L. Schilling, R. Song, and Z. Vondraček, *Bernstein Functions: Theory and Applications*, second edition, De Gruyter Studies in Mathematics 37, 2012.
- [15] R. P. Stanley, Differentiably finite power series, *European J. Combin.* **1** (1980), 175–188.